Supplementary Information

Delta Dependency Prioritizes Paralog-Based Synthetic Lethality Candidates Across Solid Tumor Types

Qingqing Mo[1,2], Tao Zhu[1,2,*]

[1] Department of Obstetrics and Gynecology, National Clinical Research Center for Obstetrics and Gynecology, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China

[2] Key Laboratory of Cancer Invasion and Metastasis (Ministry of Education), Hubei Key Laboratory of Tumor Invasion and Metastasis, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China

[*] Correspondence: zhutao@tjh.tjmu.edu.cn

Supplementary Information

Delta Dependency Prioritizes Paralog-Based Synthetic Lethality Candidates Across Solid Tumor Types

Contents

1 Supplementary Figures

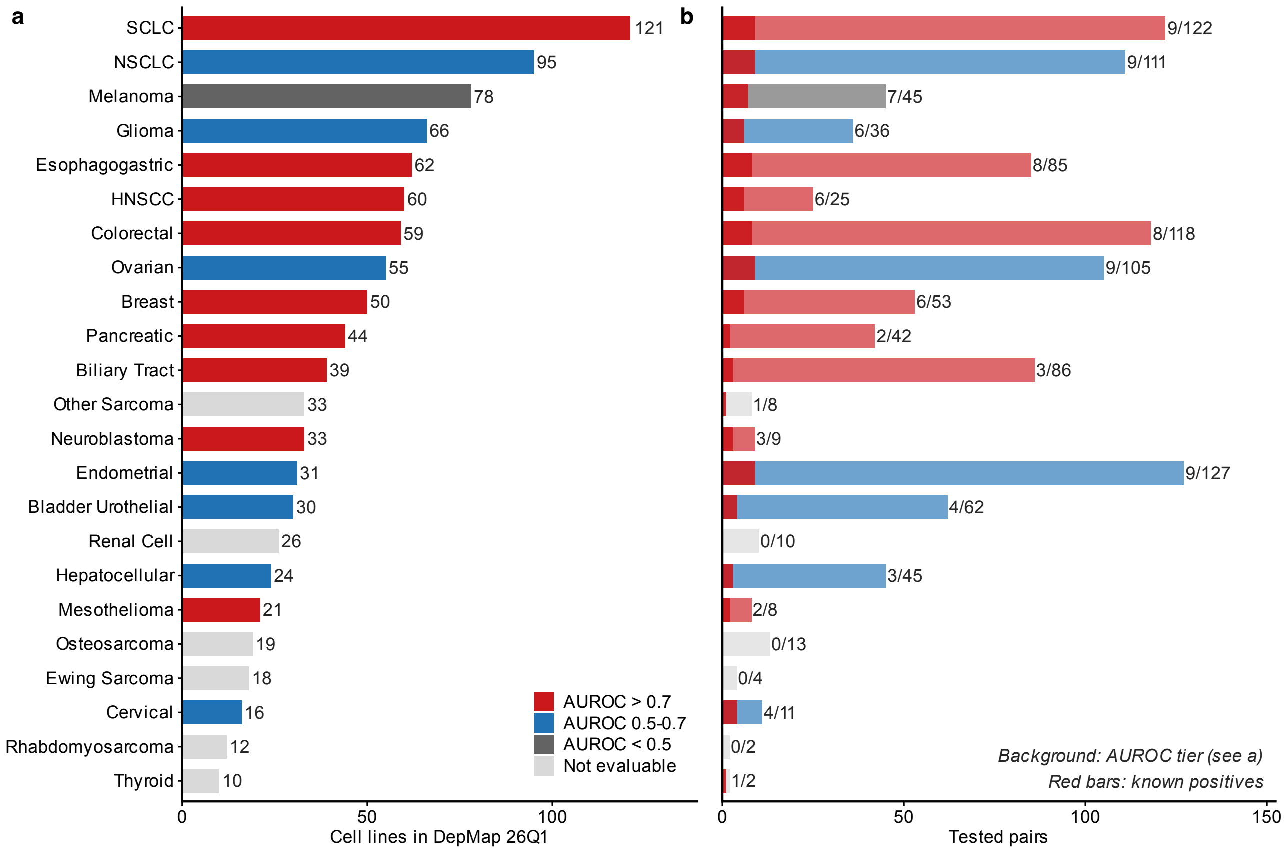

Fig. S1: Cell line landscape and evaluation summary across 23 solid tumor types. a, Cell line counts per cancer type, colored by DD AUROC tier (red: AUROC > 0.7; blue: 0.5–0.7; gray: < 0.5; light gray: not evaluable). b, Tested paralog pairs per cancer type; red bars show known gold-standard positives. Numbers indicate known/total pairs.

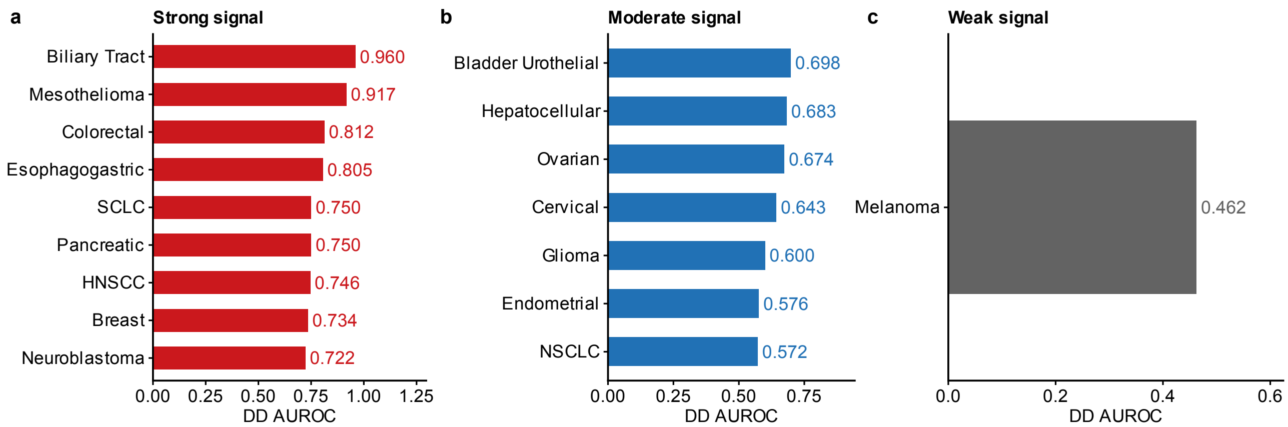

Fig. S2: Cross-cancer DD AUROC. DD AUROC for each of the 17 evaluable solid tumor types, grouped by signal strength. a, Strong signal (AUROC $\geq$ 0.7; 9 lineages including Biliary Tract 0.960, Mesothelioma 0.917, Colorectal 0.812). b, Moderate signal (AUROC 0.5–0.7; 7 lineages). c, Weak signal (AUROC < 0.5; Melanoma 0.462). Dashed line indicates random classifier (0.5). Bar labels show the number of tested paralog pairs per lineage. Cancer types with <2 known positive pairs were excluded from AUROC evaluation (see Table S1 for the 6 non-evaluable types).

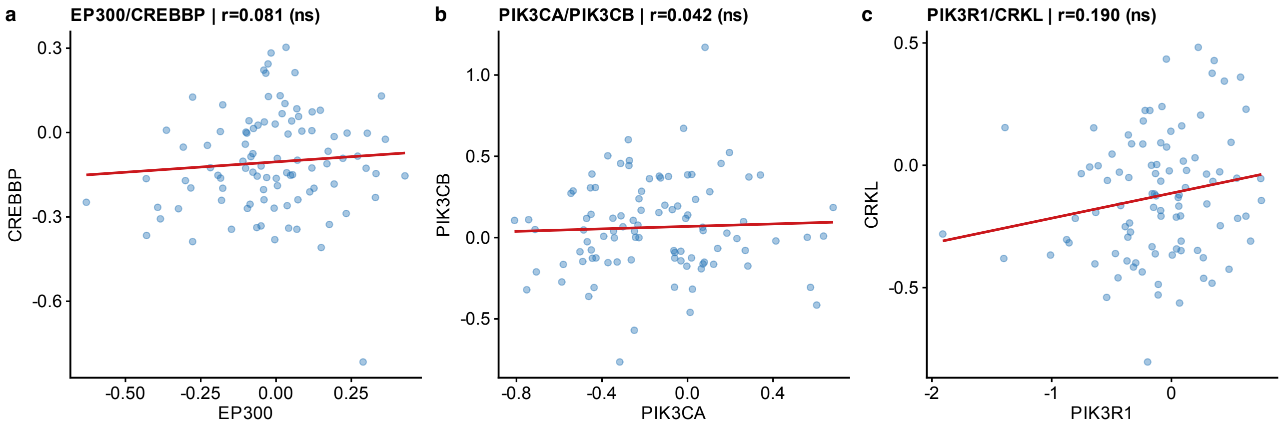

Fig. S3: CPTAC per-cohort scatter plots (UCEC). Representative scatter plot of EP300 vs. CREBBP protein abundance ($\log_2$ normalized intensity) in the UCEC cohort ($n = 83$). Each point represents one tumor sample. Line: linear regression fit with 95% confidence band. Pearson r and BH-corrected q-value shown. Similar scatter plots for other cohorts and paralog pairs are available upon request.

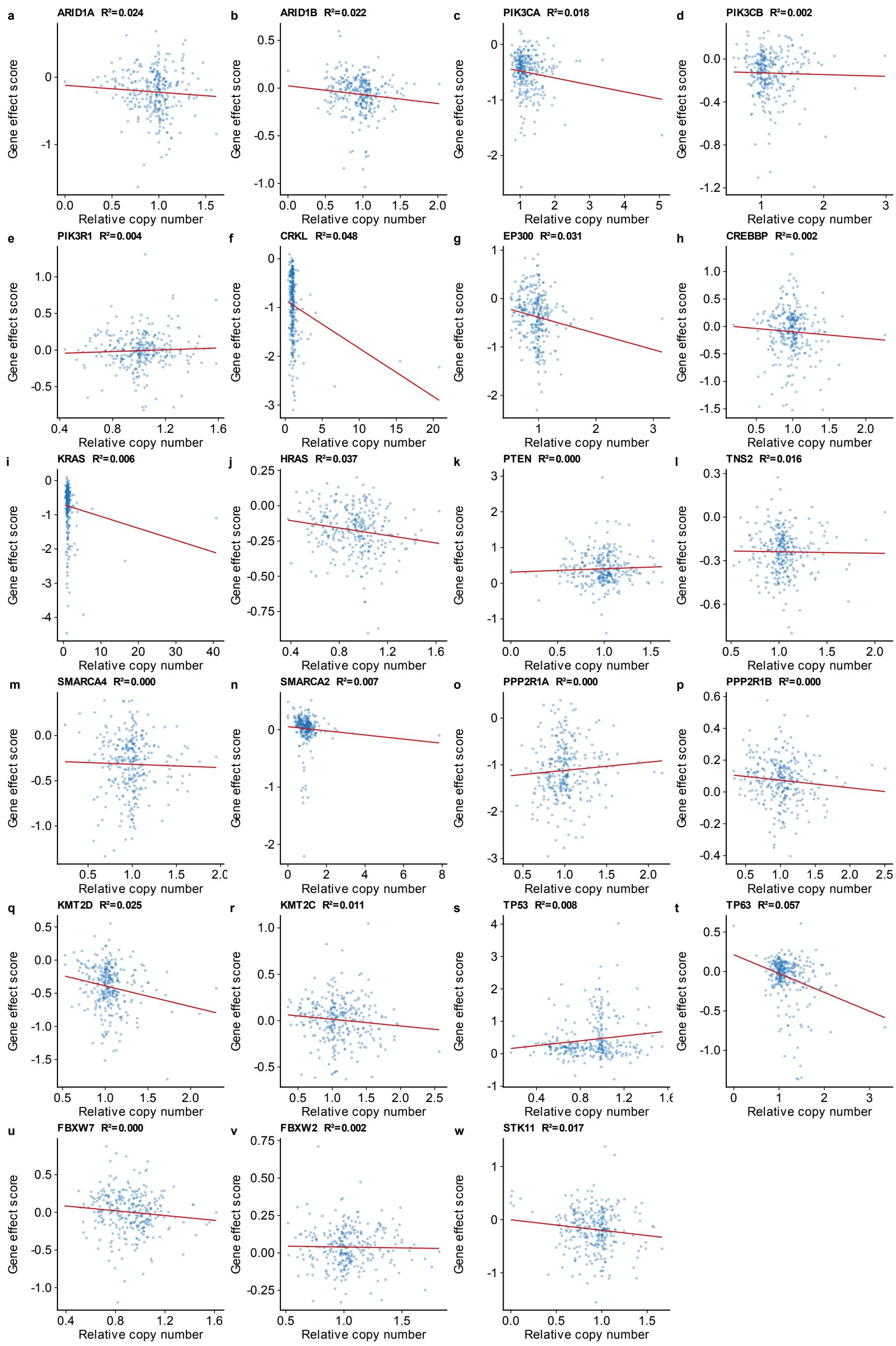

Fig. S4: CNV independence analysis. Scatter plots of relative copy number (x-axis) vs. Chronos gene-effect score (y-axis) for each of 23 paralog genes across 1,192 cell lines with both CNV and dependency data. Each panel shows one gene with the linear regression fit (red line) and R^2 value. All $R^2 < 0.10$, indicating that copy number variation explains <10% of gene-effect variance. Multivariate regression controlling for driver mutation and TP53 co-mutation left DD estimates essentially unchanged after CNV adjustment (see main text).

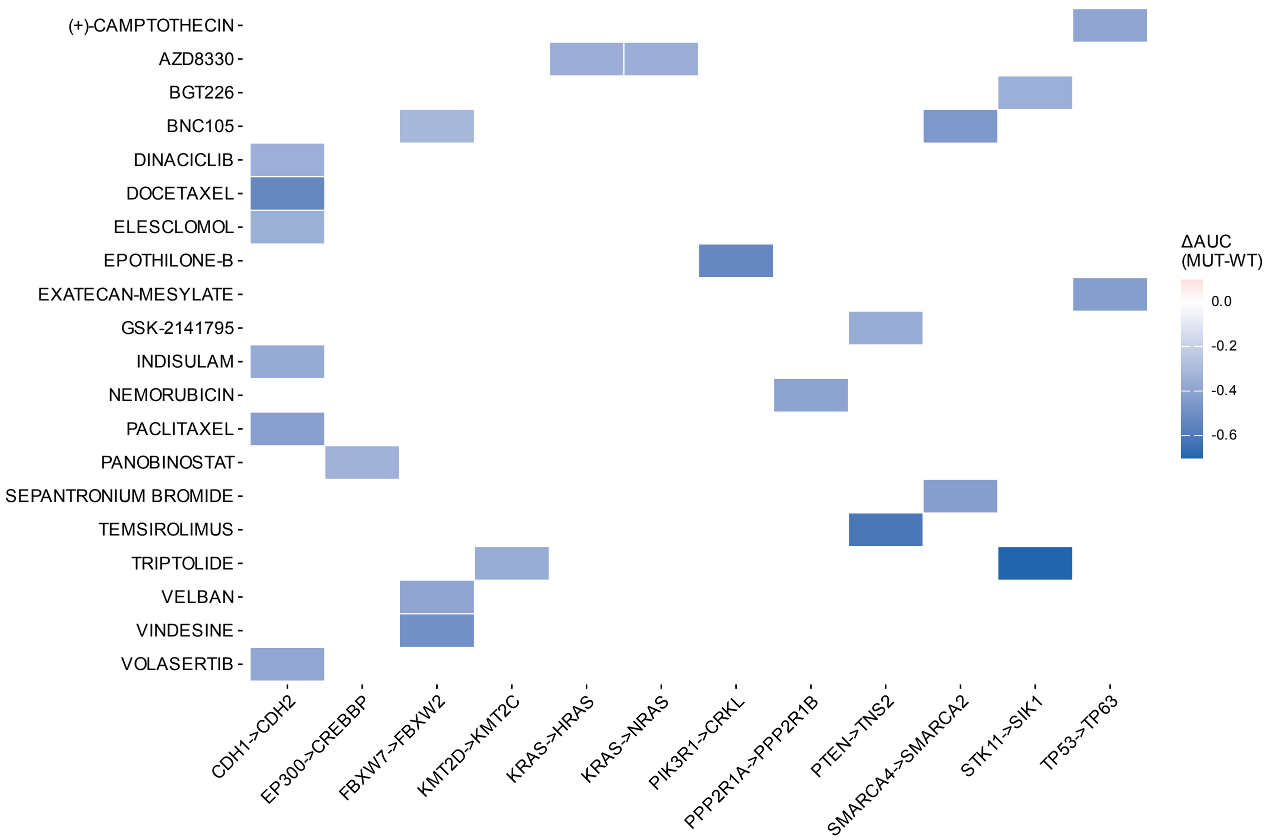

Fig. S5: PRISM drug selectivity heatmap. Heatmap of drug sensitivity differential (ΔAUC = mean AUC in driver-mutant minus mean AUC in wild-type cell lines) for the top compound–paralog pair associations from the PRISM Repurposing screen (1,482 compounds, 727 cell lines). Negative ΔAUC indicates selective sensitivity in driver-mutant cells. Associations shown met discovery-stage criteria (BH $q < 0.25$, $\Delta\text{AUC} < 0$, enrichment> 0.1). Known drug–target biology is recovered: MEK inhibitors with KRAS-mutant lines, mTOR/AKT inhibitors with PTEN-mutant lines, HDAC inhibitors with EP300-mutant lines.

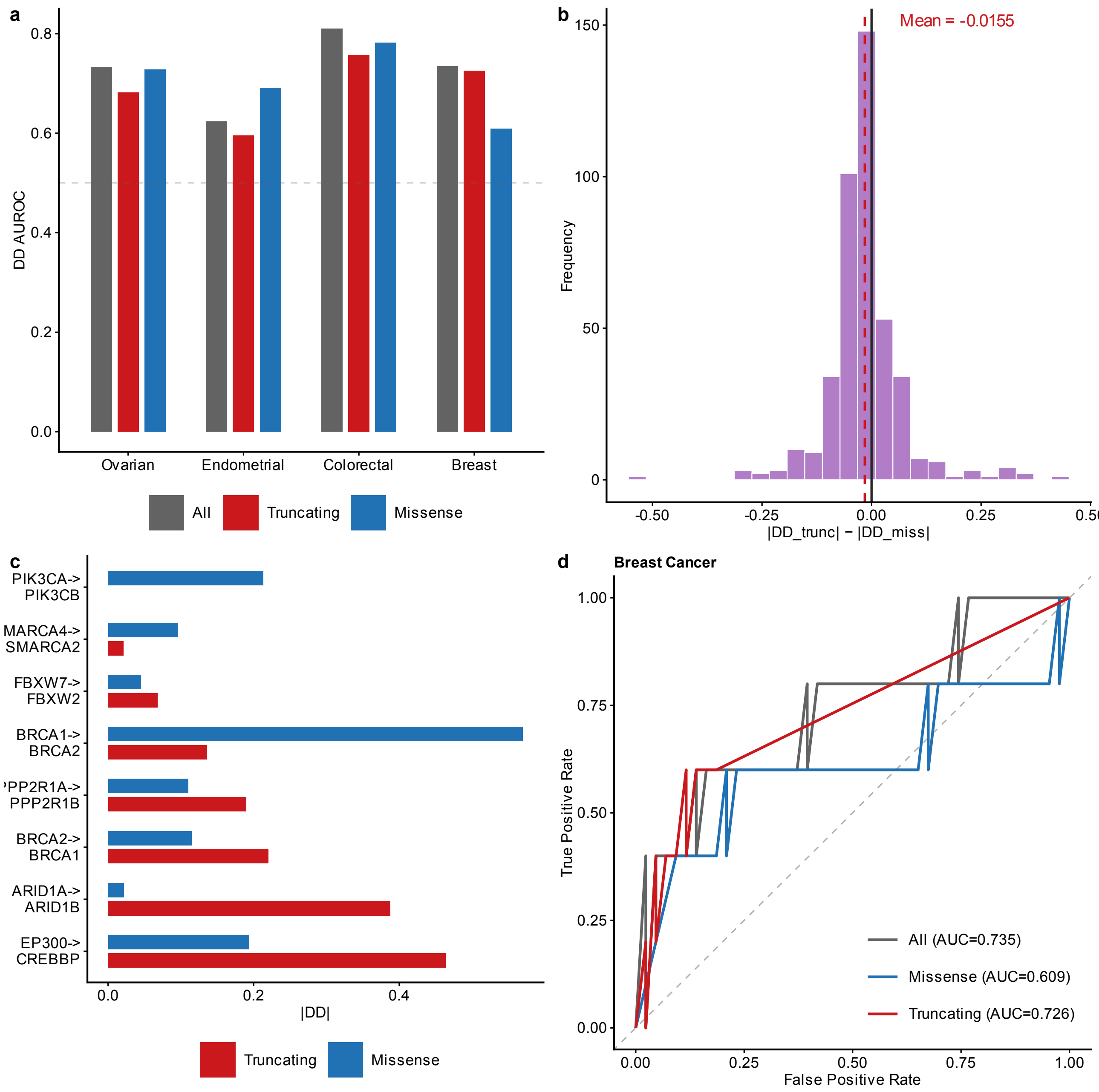

Fig. S6: Mutation-type stratification. a, DD AUROC stratified by mutation type (truncating vs. missense) across cancer types. b, Distribution of ΔDD (truncating minus missense) across all tested driver–paralog pairs; overall mean near zero (−0.016), indicating that truncating > missense is pair-specific rather than universal. c, ΔDD for known gold-standard SL pairs showing that ARID1A→ARID1B (+0.368 in Ovarian) and EP300→CREBBP (+0.314 in Colorectal) drive the trend. d, ROC curves for Breast cancer comparing truncating-only (AUROC = 0.726) vs. overall DD performance. Panel d is computed on the mutation-type analysis frame (48 Breast pairs); the resulting "All" AUC (0.735) therefore differs slightly from the cross-cancer frame value (0.734; Fig. 1a).

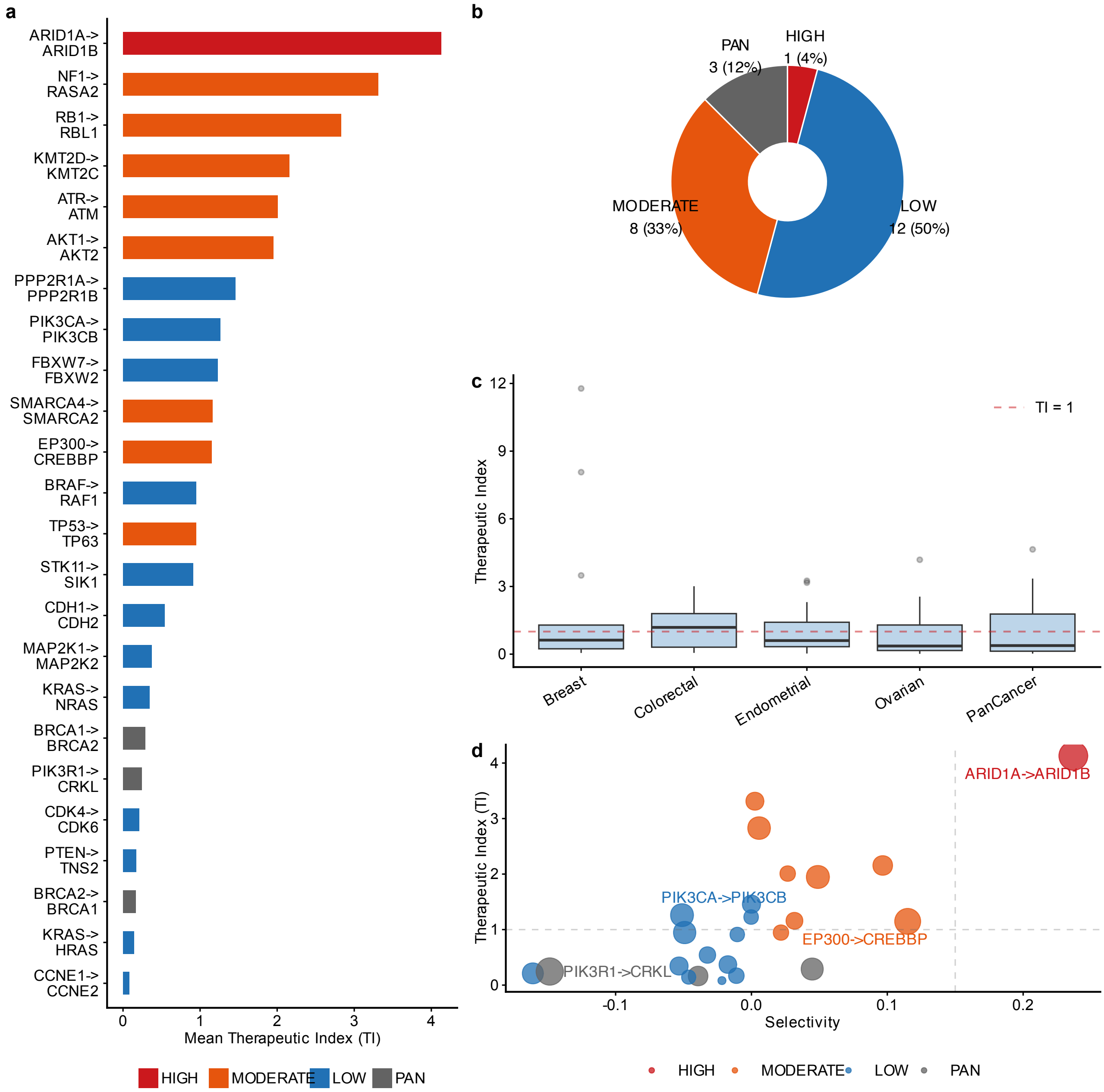

Fig. S7: Therapeutic window analysis. a, All 24 paralog pairs ranked by mean Therapeutic Index (TI) across 5 cancer contexts (Ovarian, Endometrial, Breast, Colorectal, PanCancer). ARID1A→ARID1B ranks first (TI = 4.13). b, In vitro selectivity tier classification: HIGH_SELECTIVITY (selectivity > 0.15 and TI > 1.0), MODERATE, LOW_SELECTIVITY, and PAN_ESSENTIAL ($f_{\text{pan-essential}} > 0.5$). c, TI values broken down by individual cancer context showing cross-lineage consistency for top candidates. d, Selectivity vs. TI scatter plot; ARID1A→ARID1B is the only pair in the upper-right quadrant (high TI and high selectivity).

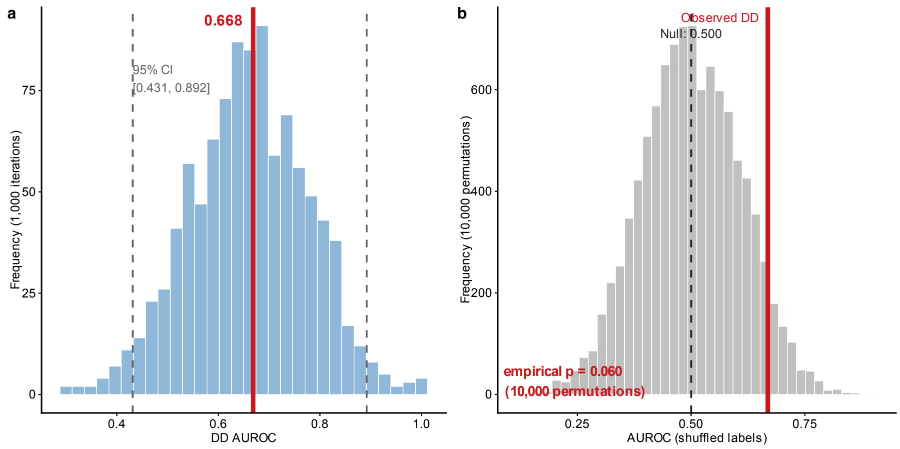

Fig. S8: Bootstrap and negative control (per-pair evaluation, 77 pairs, 8 known positives). a, Bootstrap distribution (1,000 iterations; 95% CI: 0.431–0.892). b, Null distribution from 10,000 label-shuffled permutations (empirical $p = 0.060$). The observed per-pair AUROC (0.668) exceeds the null mean (0.500) but does not reach conventional significance, reflecting limited power with only 8 positive pairs. The cancer-type-specific AUROC (0.794, reported in the benchmark comparison) is higher because it evaluates each driver×paralog×lineage combination separately rather than aggregating across lineages.

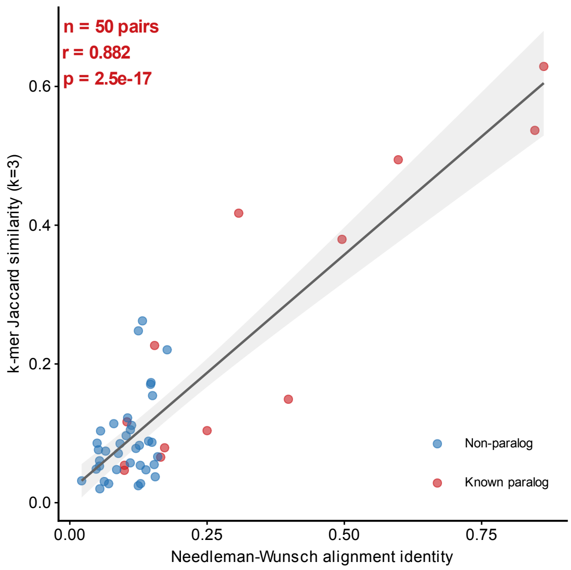

Fig. S9: k-mer Jaccard vs. Needleman-Wunsch alignment identity validation. Scatter plot of k-mer Jaccard similarity ($k = 3$) vs. global alignment identity for 50 protein pairs (13 known paralogs, 37 non-paralog controls). Red: known paralog-SL pairs; Blue: non-paralog pairs. Gray line: linear regression with 95% CI.

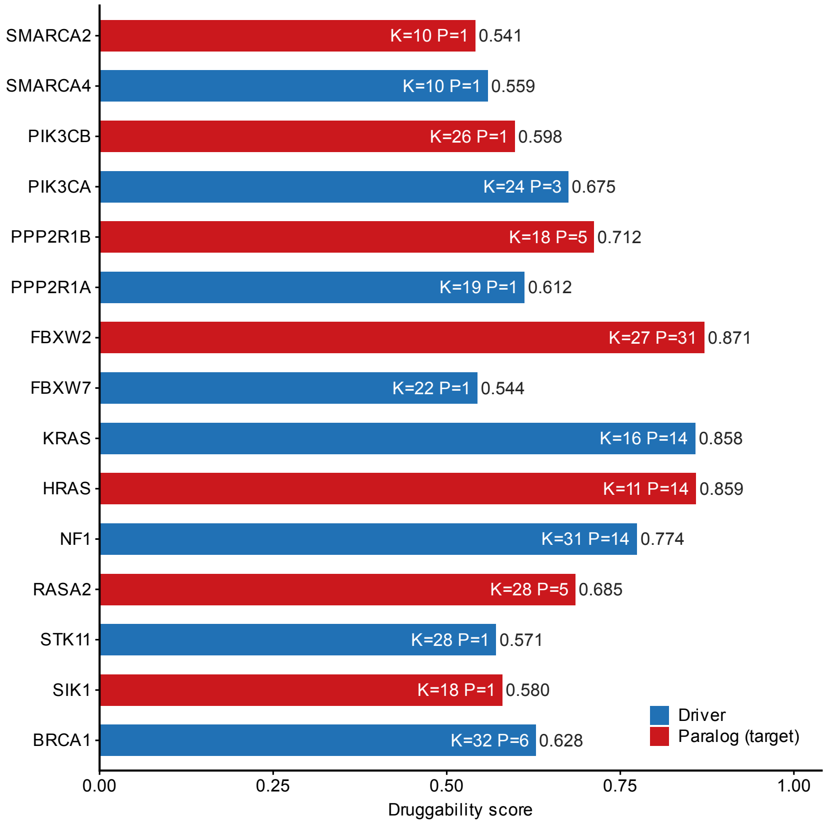

Fig. S10: Structure-based druggability assessment. Composite druggability score for 15 proteins (7 paralog targets in red, 8 driver genes in blue) based on ESMFold-predicted structures. Scores integrate structural confidence (pLDDT), hydrophobic content, lysine density (K, relevant for PROTAC conjugation), and predicted pocket count (P). Structures were limited to 400 residues per protein; ARID1A/ARID1B structures were not available from AlphaFold DB and are assessed by sequence-level features only (main text Table 2). FBXW2 (0.871) and KRAS/HRAS (0.858–0.859) show the highest scores due to well-folded compact domains.

2 Supplementary Tables

2.1 Table S1: Cell line and evaluation summary across 23 solid tumor types

Label	Full name	Cell Lines	Pairs	Known+	DD AUROC
Biliary Tract	Biliary tract cancer	39	86	3	0.960
Mesothelioma	Pleural mesothelioma	21	8	2	0.917
Colorectal	Colorectal adenocarcinoma	59	118	8	0.812
Esophagogastric	Esophagogastric adenocarcinoma / esophageal SCC	62	85	8	0.805
SCLC	Small cell lung cancer	121	122	9	0.750
Pancreatic	Pancreatic adenocarcinoma	44	42	2	0.750
HNSCC	Head and neck squamous cell carcinoma	60	25	6	0.746
Breast	Invasive breast carcinoma	50	53	6	0.734
Neuroblastoma	Neuroblastoma	33	9	3	0.722
Bladder Urothelial	Bladder urothelial carcinoma	30	62	4	0.698
Hepatocellular	Hepatocellular carcinoma	24	45	3	0.683
Ovarian	Ovarian epithelial carcinoma	55	105	9	0.674
Cervical	Cervical carcinoma	16	11	4	0.643
Glioma	Diffuse glioma	66	36	6	0.600
Endometrial	Endometrial carcinoma	31	127	9	0.576
NSCLC	Non-small cell lung cancer	95	111	9	0.572
Melanoma	Melanoma	78	45	7	0.462
Renal Cell	Renal cell carcinoma	26	10	0	—
Osteosarcoma	Osteosarcoma	19	13	0	—
Ewing Sarcoma	Ewing sarcoma	18	4	0	—
Other Sarcoma	Other soft-tissue sarcoma	33	8	1	—
Rhabdomyosarcoma	Rhabdomyosarcoma	12	2	0	—
Thyroid	Thyroid carcinoma	10	2	1	—

Table S1: Cell line counts and DD AUROC across all 23 analyzed solid tumor types. Top: AUROC > 0.7 (9 types). Middle: AUROC ≤ 0.7 (8 types). Bottom: insufficient known positives for AUROC (6 types). Known+: number of gold-standard positive pairs evaluable in each lineage. "—": AUROC not computable (<2 known positives). Full name: standardized cancer-type name corresponding to each short label; SCC, squamous cell carcinoma.

2.2 Table S2: Complete paralog-SL pair analysis results

The full analysis results are provided as a tab-separated file: `TableS2_FullResults.tsv` (118 rows × 13 columns, representing all driver×paralog×cancer-type combinations passing the minimum sample-size filter).

Data dictionary:

- `driver_gene`: Driver gene with damaging mutation.
- `paralog_gene`: Candidate paralog tested for conditional dependency.
- `cancer_type`: Solid tumor lineage (DepMap OncotreePrimaryDisease).
- `pcs`: Paralog Compensation Score (DD × necessity).
- `delta_expression`: ΔExpression between MUT and WT groups.
- `necessity`: Mean gene-effect score of the paralog (more negative = more essential).
- `dependency_dd`: Delta Dependency (signed; positive = greater dependency in MUT).
- `composite_score`: Weighted composite of PCS, DD, and ΔExpression.
- `novelty`: "Known" if in gold-standard set (Table S3), else "Novel".

- `q_value`: Benjamini-Hochberg adjusted p-value within each driver gene.
- `mutation_frequency`: Fraction of cell lines with driver mutation in this lineage.
- `n_mut`, `n_wt`: Number of mutant and wild-type cell lines.
- `is_known_paralog_sl`: Boolean; TRUE if pair is in gold-standard set.

2.3 Table S3: Gold-standard paralog-SL pairs with evidence provenance

Driver	Paralog	Type	Evidence	Key Ref	Indep.
SMARCA4	SMARCA2	Seq. paralog	CRISPR screen	Hoffman 2014	Yes
ARID1A	ARID1B	Seq. paralog	CRISPR + drug	Helming 2014	Yes
BRCA1	BRCA2	Func. analog	Genetic screen	Bryant 2005	Yes
EP300	CREBBP	Seq. paralog	shRNA screen	Ogiwara 2016	Yes
PIK3CA	PIK3CB	Seq. paralog	Drug + KO	Wee 2008	Yes
AKT1	AKT2	Seq. paralog	siRNA viability	Literature	Partial
STK11	SIK1	Partial hom.	Genetic screen	Literature	Partial
FBXW7	FBXW2	Seq. paralog	DepMap analysis	DepMap	No
PPP2R1A	PPP2R1B	Seq. paralog	DepMap analysis	DepMap	No
CCNE1	CCNE2	Seq. paralog	Genetic screen	Literature	Yes
CDK4	CDK6	Seq. paralog	Drug (palbociclib)	Clinical	Yes
MAP2K1	MAP2K2	Seq. paralog	Drug (trametinib)	Clinical	Yes

Table S3: Twelve gold-standard pairs with evidence provenance. Type: Seq. paralog (shared gene family, conserved domains), Func. analog (shared pathway role, no sequence homology), Partial hom. (same superfamily, divergent). Indep.: whether evidence is independent of DepMap CRISPR data. "No" indicates pairs whose SL status was originally identified from DepMap-derived analyses, creating potential circularity.

2.4 Table S4: Pan-cancer summary statistics

Cancer Type	Cell Lines	Pairs	DD AUROC
Biliary Tract	39	86	0.960
Mesothelioma	21	8	0.917
Colorectal	59	118	0.812
Esophagogastric	62	85	0.805
SCLC	121	122	0.750
Pancreatic	44	42	0.750
HNSCC	60	25	0.746
Breast	50	53	0.734
Neuroblastoma	33	9	0.722
Bladder Urothelial	30	62	0.698
Hepatocellular	24	45	0.683
Ovarian	55	105	0.674
Cervical	16	11	0.643
Glioma	66	36	0.600
Endometrial	31	127	0.576
NSCLC	95	111	0.572
Melanoma	78	45	0.462

Table S4: DD AUROC across all 17 evaluable solid tumor types (sorted by AUROC). Horizontal line separates AUROC > 0.7 (top) from AUROC ≤ 0.7 (bottom). Full lineage names are given in Table S1.

2.5 Table S5: 40-gene driver panel

Gene	Cancer Types	Source	Category
TP53	Pan-cancer	TCGA+COSMIC	TSG
KRAS	Lung, CRC, Pancreatic	TCGA+COSMIC	Oncogene
PIK3CA	Breast, Endometrial, Cervical	TCGA+COSMIC	Oncogene
PTEN	Endometrial, Ovarian, Breast	TCGA+COSMIC	TSG
ARID1A	Endometrial, Ovarian, CRC	TCGA+COSMIC	TSG
BRCA1	Ovarian, Breast	TCGA+COSMIC	TSG
BRCA2	Ovarian, Breast	TCGA+COSMIC	TSG
BRAF	Melanoma, CRC, Thyroid	TCGA+COSMIC	Oncogene
NF1	Glioma, Breast, Lung	TCGA+COSMIC	TSG
RB1	SCLC, Breast, Bladder	TCGA+COSMIC	TSG

Table S5: First 10 of 40 driver genes (complete list in TableS5_DriverGenes.tsv). Selection criteria: recurrently mutated in $\geq$1 TCGA pan-cancer study or COSMIC Cancer Gene Census, with $\geq$3 mutant cell lines in at least one DepMap lineage.

2.6 Table S6: All 24 paralog pairs ranked by preclinical prioritization score

Driver	Paralog	—DD—	TI	Select.	Class	Type
ARID1A	ARID1B	0.267	4.130	0.237	HS	Seq
NF1	RASA2	0.066	3.313	0.003	MOD	Seq
RB1	RBL1	0.134	2.829	0.006	MOD	Seq
KMT2D	KMT2C	0.087	2.155	0.097	MOD	Seq
ATR	ATM	0.040	2.009	0.027	MOD	Seq
AKT1	AKT2	0.141	1.948	0.049	MOD	Seq
PPP2R1A	PPP2R1B	0.068	1.458	0.000	LS	Seq
PIK3CA	PIK3CB	0.138	1.261	−0.051	LS	Seq
FBXW7	FBXW2	0.032	1.226	0.000	LS	Seq
SMARCA4	SMARCA2	0.053	1.159	0.032	MOD	Seq
EP300	CREBBP	0.200	1.148	0.115	MOD	Seq
BRAF	RAF1	0.129	0.947	−0.049	LS	Seq
TP53	TP63	0.039	0.946	0.022	MOD	Seq
STK11	SIK1	0.031	0.912	−0.010	LS	Part
CDH1	CDH2	0.049	0.542	−0.032	LS	Seq
MAP2K1	MAP2K2	0.062	0.370	−0.017	LS	Seq
KRAS	NRAS	0.068	0.344	−0.053	LS	Seq
BRCA1	BRCA2	0.125	0.289	0.045	PE	Func
PIK3R1	CRKL	0.231	0.245	−0.148	PE	Seq
CDK4	CDK6	0.107	0.212	−0.161	LS	Seq
PTEN	TNS2	0.041	0.170	−0.011	LS	Seq
BRCA2	BRCA1	0.087	0.164	−0.039	PE	Func
KRAS	HRAS	0.029	0.144	−0.046	LS	Seq
CCNE1	CCNE2	0.006	0.081	−0.022	LS	Seq

Table S6: All 24 analyzed paralog pairs ranked by mean Therapeutic Index (TI). —DD—: mean absolute Delta Dependency across cancer contexts. Select.: mean selectivity (fraction mutant-essential minus fraction wild-type-essential). Class: HS=HIGH_SELECTIVITY, MOD=MODERATE, LS=LOW_SELECTIVITY, PE=PAN_ESSENTIAL. Type: Seq=sequence paralog, Part=partial homology, Func=functional analog.